T3 UV one-loop RGE term comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$

R_W	$16\pi^2\beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \to g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$

Quantity: g_1

$$16\pi^2\,\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$$

Column	Representative term structure
T1	$\frac{25{g_1}^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\frac{25{g_1}^3}{2}$

$\beta_{g_2}^{(1)}$

Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu} = \beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{17{g_2}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-\frac{17{g_2}^3}{6}$

$\beta_{g_3}^{(1)}$

Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu} = \beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7{g_3}^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-7{g_3}^3$

$\beta^{(1)}_{Y_u}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$
T2	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$
T3	$-8{g_3}^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

$\beta^{(1)}_{Y_d}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

Terms 1 of 3

Column	Representative term structure
T1	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$
T3	$-8{g_3}^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

Quantity: Y_e

$$16\pi^2\,\mu\frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$
T3	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$
T3	$-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$

$\beta_{y_1}^{(1)}$

Quantity: y_1

$$16\pi^2\,\mu\frac{dy_1}{d\mu} = \beta_{y_1}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$T_\nu^{(1)}\,(y_1)_{ij}$
T2	$+\frac{1}{2}(\left(y_2\;(y_2)^\dagger\;y_1\right)_{ij} + \left(Y_e\;(Y_e)^\dagger\;y_1\right)_{ij} + 3\left(y_1\;(y_1)^\dagger\;y_1\right)_{ij})$
T3	$-\frac{9}{4}{g_2}^2\,(y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$T_\nu^{(1)}\,(y_1)_{ij}$	$+\frac{1}{2}(\left(y_2\;(y_2)^\dagger\;y_1\right)_{ij} + \left(Y_e\;(Y_e)^\dagger\;y_1\right)_{ij} + 3\left(y_1\;(y_1)^\dagger\;y_1\right)_{ij})$	$-\frac{9}{4}{g_2}^2\,(y_1)_{ij}$

Quantity: y_1

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{15}{4} {g_1}^2 (y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-\frac{15}{4} {g_1}^2 (y_1)_{ij}$

$\beta_{y_2}^{(1)}$

Quantity: y_2

$$16\pi^2\,\mu\frac{dy_2}{d\mu} = \beta_{y_2}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$T_\nu^{(2)}\,(y_2)_{ij}$
T2	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$
T3	$-\frac{9}{4}{g_2}^2\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$T_\nu^{(2)}\,(y_2)_{ij}$	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$	$-\frac{9}{4}{g_2}^2\,(y_2)_{ij}$

Quantity: y_2

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{15}{4} {g_1}^2 (y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-\frac{15}{4} {g_1}^2 (y_2)_{ij}$

$\beta^{(1)}_{M_F}$

Quantity: M_F

$$16\pi^2\,\mu\,\frac{dM_F}{d\mu} = \beta^{(1)}_{M_F}$$

Column	Representative term structure
T1	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$
T2	$+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$
T3	$-6{g_1}^2\,(M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$	$+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$	$-6{g_1}^2\,(M_F)_{ij}$

$\beta^{(1)}_{m_1^2}$

Quantity: m_1^2

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

Terms 1 of 2

Column	Representative term structure
T1	$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$
T2	$-4\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$
T3	$-\frac{9{g_2}^2 m_1^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	$-4\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$	$-\frac{9{g_2}^2 m_1^2}{2}$

Quantity: m_1^2

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{27{g_1}^2 m_1^2}{2}$
T2	$+4\lambda_{12}^{(1)} m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-\frac{27{g_1}^2 m_1^2}{2}$	$+4\lambda_{12}^{(1)} m_2^2$

$\beta^{(1)}_{m_2^2}$

Quantity: m_2^2

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu} = \beta^{(1)}_{m_2^2}$$

Terms 1 of 2

Column	Representative term structure
T1	$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$
T2	$-4\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$
T3	$-\frac{9{g_2}^2 m_2^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	$-4\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$	$-\frac{9{g_2}^2 m_2^2}{2}$

Quantity: m_2^2

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{3{g_1}^2 m_2^2}{2}$
T2	$+4\lambda_{12}^{(1)} m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-\frac{3{g_1}^2 m_2^2}{2}$	$+4\lambda_{12}^{(1)} m_1^2$

$\beta^{(1)}_{\lambda_1}$

Quantity: λ_1

$$16\pi^2\,\mu\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_1 (12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$
T3	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\lambda_1 (12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Quantity: λ_1

Terms 2 of 5

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$
T2	$+4\lambda_5\ (\lambda_5)^*$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+4\lambda_5\ (\lambda_5)^*$	$+\frac{9{g_2}^4}{4}$

Quantity: λ_1

Terms 3 of 5

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{{3g_2}^2}{2}-3\lambda_1\right)$
T2	$-9{g_2}^2\lambda_1$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+{g_1}^2\left(\frac{{3g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$	$+\frac{3{g_1}^4}{4}$

Quantity: λ_1

Terms 4 of 5

Column	Representative term structure
T1	$+12{\lambda_1}^2$
T2	$+4{\lambda_{HS_1}^{(1)}}^2$
T3	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$

Quantity: λ_1

Terms 5 of 5

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)\ 2}$
T2	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+4\lambda_{HS_2}^{(1)\ 2}$	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$

$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$

Quantity: $\lambda^{(1)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

Terms 1 of 4

Column	Representative term structure
T1	$4\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$
T2	$-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$4\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$	$-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	$+\frac{9{g_2}^4}{4}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$
T2	$-9{g_2}^2\lambda_{S_1}^{(1)}$
T3	$+\frac{243{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{243{g_1}^4}{4}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+4{\lambda_{12}^{(1)}}^2$
T2	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
T3	$+4{\lambda_{HS_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+4{\lambda_{12}^{(1)}}^2$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+4{\lambda_{HS_1}^{(1)}}^2$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
T2	$+12{\lambda_{S_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$	$+12{\lambda_{S_1}^{(1)}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

Quantity: $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$4\lambda^{(1)}_{S_2}T^{(2)}_{\nu}$
T2	$-4\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$4\lambda^{(1)}_{S_2}T^{(2)}_{\nu}$	$-4\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$+\frac{9{g_2}^4}{4}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T2	$-9{g_2}^2\lambda_{S_2}^{(1)}$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)\,2}$
T2	$+\frac{\lambda_{12}^{(A)\,2}}{4}$
T3	$+4\lambda_{HS_2}^{(1)\;\;2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+4\lambda_{12}^{(1)\,2}$	$+\frac{\lambda_{12}^{(A)\,2}}{4}$	$+4\lambda_{HS_2}^{(1)\;\;2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
T2	$+12{\lambda_{S_2}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$	$+12{\lambda_{S_2}^{(1)}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

Quantity: $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+6\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+6\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
T3	$+\frac{27{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_1}$	$+\frac{27{g_1}^4}{4}$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 3 of 4

Column	Representative term structure
T1	$-15{g_1}^2\lambda^{(1)}_{HS_1}$
T2	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$
T3	$+6\lambda_1\lambda^{(1)}_{HS_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-15{g_1}^2\lambda^{(1)}_{HS_1}$	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	$+6\lambda_1\lambda^{(1)}_{HS_1}$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 4 of 4

Column	Representative term structure
T1	$+4{\lambda^{(1)}_{HS_1}}^{2}$
T2	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$
T3	$+\frac{3{\lambda^{(A)}_{HS_1}}^{2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	$+\frac{3{\lambda^{(A)}_{HS_1}}^{2}}{4}$

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

Quantity: $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+6\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+6\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$-9{g_2}^2\lambda^{(1)}_{HS_2}$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{g_1}^4}{4}$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 3 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda^{(1)}_{HS_2}$
T2	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$
T3	$+6\lambda_1\lambda^{(1)}_{HS_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-3{g_1}^2\lambda^{(1)}_{HS_2}$	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	$+6\lambda_1\lambda^{(1)}_{HS_2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+4{\lambda_{HS_2}^{(1)}}^2$
T2	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+4{\lambda_{HS_2}^{(1)}}^2$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

Quantity: $\lambda^{(1)}_{12}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(1)}_{12}(2T^{(1)}_{\nu} + 2T^{(2)}_{\nu})$
T2	$-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$+3\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\lambda^{(1)}_{12}(2T^{(1)}_{\nu} + 2T^{(2)}_{\nu})$	$-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$+3\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$-9{g_2}^2\lambda_{12}^{(1)}$
T3	$+\frac{27{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{12}^{(1)}$	$+\frac{27{g_1}^4}{4}$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$-15{g_1}^2\lambda_{12}^{(1)}$
T2	$+4{\lambda_{12}^{(1)}}^2$
T3	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-15{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta^{(1)}_{\lambda_5}$

Quantity: λ_5

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu} = \beta^{(1)}_{\lambda_5}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_5\left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_5$
T3	$-9{g_1}^2\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\lambda_5\left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_5$	$-9{g_1}^2\lambda_5$

Quantity: λ_5

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+4\lambda_{HS_2}^{(1)}\lambda_5$

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(A)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(A)}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$
T2	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{12}^{(A)}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

Quantity: $\lambda^{(A)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$+{g_1}^2\left(-18{g_2}^2-15\lambda_{HS_1}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
T3	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+{g_1}^2\left(-18{g_2}^2-15\lambda_{HS_1}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Quantity: $\lambda^{(A)}_{HS_1}$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_1\lambda^{(A)}_{HS_1}$
T2	$+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$
T3	$+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+2\lambda_1\lambda^{(A)}_{HS_1}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$	$+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

Quantity: $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+16\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+16\lambda_5\ (\lambda_5)^*$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
T3	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Quantity: $\lambda^{(A)}_{HS_2}$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_1\lambda^{(A)}_{HS_2}$
T2	$+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$
T3	$+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+2\lambda_1\lambda^{(A)}_{HS_2}$	$+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$	$+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$